

Áron Imre Németh

MSc Student in Mechatronics & Robotics



CONTACT

Munich, Germany

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AVAILABILITY

Immediately

20–24 hours / week

Alongside my master's thesis

TECHNICAL SKILLS

Programming

Python C++ C CUDA

MATLAB

Machine learning

PyTorch scikit-learn

pandas NumPy

Robotics & control

ROS 2 Isaac Lab

Isaac Sim Franka RViz

MPC / MPPI

Tools

Git Docker Linux

LaTeX

LANGUAGES

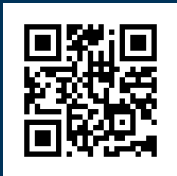
Hungarian (Native)

German (C1)

English (C1)

Personal Website

PORTFOLIO



Robotics MSc student at TUM: learning-based manipulation, control and ROS 2.

RESEARCH & EXPERIENCE

Research Intern

05/2026–present



Learning Systems and Robotics Lab — TU Munich

- Built sampling-based MPC (MPPI) with Conditional Flow Matching priors for Franka grasping and LEGO pick-and-place in Isaac Lab.
- Used FoundationPose for 6D pose estimation and tracking from wrist-camera images.
- Simulation: 91.7% grasp-and-lift (550/600, six objects, ground-truth poses); 92.4% pick-and-place (231/250, camera-estimated poses).
- Transferring to Franka Research 3 with ROS 2 and CRISP/ros2_control joint-impedance control at 1 kHz.

Sampling-based MPC Flow Matching FoundationPose Isaac Lab

Data Analysis & AI Intern

06–10/2023

Antra ID Kft. — Budaörs

Data preprocessing and dimensionality reduction using pandas, scikit-learn, PCA, SVD and matrix factorization.

SELECTED PROJECTS

Autonomous driving in ROS 2

TU Munich — Introduction to ROS — Summer 2026

Implemented sampling-based MPC, a decision-making state machine and lane detection in a team-built Unity driving stack.

C++ ROS 2 Vehicle control

Neural raycasting for tactile simulation

TU Munich — Advanced Deep Learning for Robotics

Replaced mesh raycasting with a learned directional-distance network: approximately 32× faster inference and 93.84% ray-distance accuracy across ten objects.

Deep Learning Tactile simulation GPU inference

EDUCATION

MSc Mechatronics & Robotics

04/2025–present



Technical University of Munich

GPA: 1.9 (April 2026).

Mechanical Engineering — Erasmus

06–09/2024



Technical University of Munich — Master's exchange

Control Theory and Machine Learning — Grade: 2.0.

Master's studies in Mechatronics

02/2024–03/2025



Budapest University of Technology and Economics

Studies before transfer to TUM.

BSc Mechatronics

09/2020–01/2024



Budapest University of Technology and Economics

Biomechatronics — GPA: 4.62/5.00.

Thesis: CNN-based estimation from footstep audio. Grade: 5.

AWARD

1st place — Scientific Student Circle Conference

BME, 2023 — Anthropometric estimation from footstep recordings.